Slopes of lines

1 Define the slope of a line.

2 What is the change in the vertical distance along a line called?

3 What is the change in the horizontal distance along a line called?

4 Explain what it means if the slope of a line is zero.

5 Explain what it means if the slope of a line is undefined.

6 If we have two points and the slope formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)? Explain.

7 Using $m = \dfrac{\text{rise}}{\text{run}}$, find the slope of the line that passes through:

a Point A $(-1, 0)$ and Point B $(0, 3)$

b Point A $(2, -6)$ and the origin

8 Using $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, find the slope of the line that passes through:

a Point A $(3, 5)$ and Point B $(1, 8)$

b Point A $(-3, -1)$ and Point B $(-5, 1)$

9 Find the slope of the line that passes through the points:

a $(3, 4)$ and $(5, 4)$

b $(-3, 4)$ and $(1, 4)$.

10 Consider the points $A\,(24, -56)$, $B\,(-6, -2)$ and $C\,(-11, 7)$:

a Find the slope of AB.

b Find the slope of BC.

c Do the points A, B and C lie on a single straight line?

11 A line passing through the points $(5, 3)$ and $(2, t)$ has a slope equal to -4.
Find the value of t.

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12 A line passes through the points $(11, c)$ and $(-20, 16)$ and has a slope of $-\frac{4}{7}$. Find the value of c.

13 The distance between points (x_1, y_1) and (x_2, y_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. This formula has been used as shown: $d = \sqrt{(3-6)^2 + (7-4)^2}$. Determine the slope of the line through AB.

14 What is the slope of the segment joining Point A $(1, -1)$ and Point B $(-1, -2)$?

15 The slope of a line is 7 and A (9, 7) is a point on the line. Another point, B, on the line has $y = 56$. What is the x-value at this point, as denoted by d?

16 For each of following pairs of coordinates, determine whether it will have an undefined or defined slope .

a $(-10, 5)$ and $(-10, 12)$

b $(10, 5)$ and $(10, 1)$

c $(10, -1)$ and $(-10, -1)$

d $(-10, 5)$ and $(10, 5)$

e $(10, 7)$ and $(10, 2)$

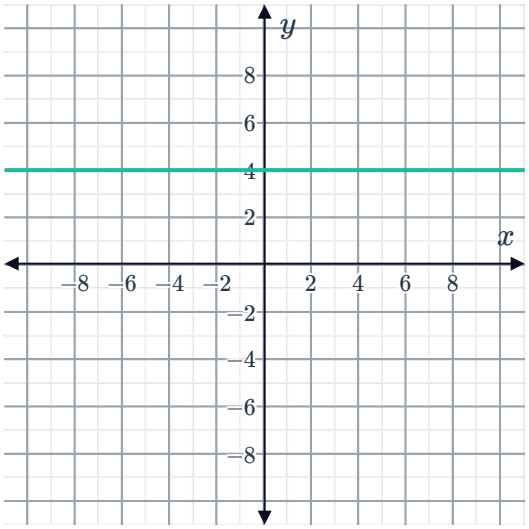
f $(10, -2)$ and $(-10, -2)$

g $(-10, 7)$ and $(-10, 12)$

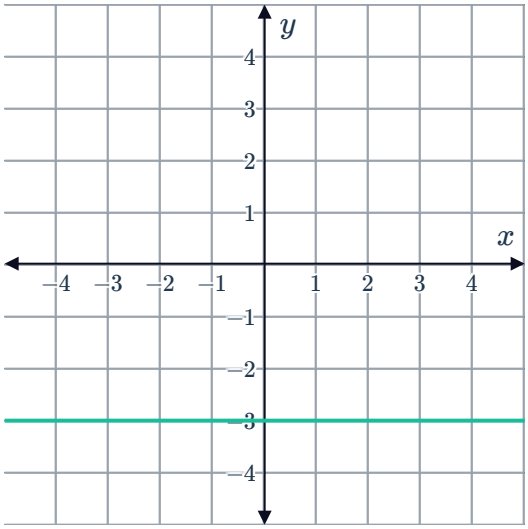
h $(-10, 7)$ and $(10, 7)$

17 Examine the graph and determine the slope of the line.

a

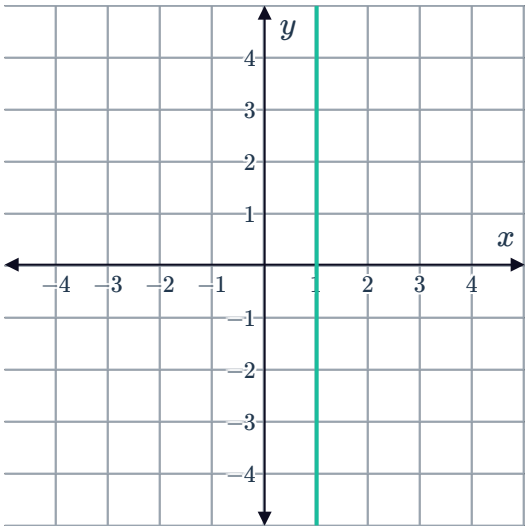


b



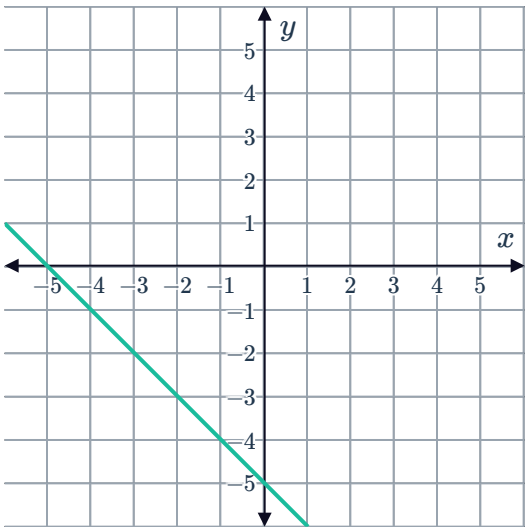
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C

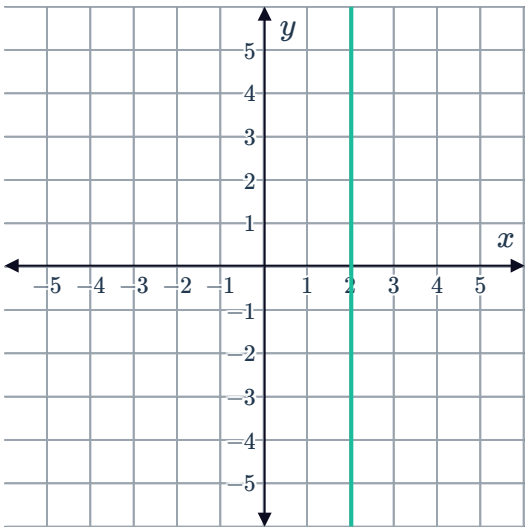


18 Consider the following graphs.

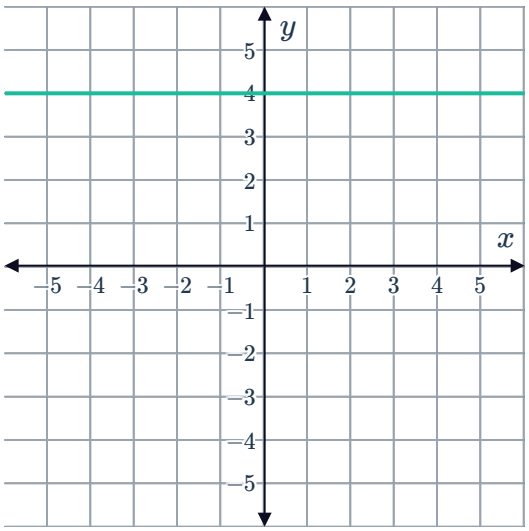
A



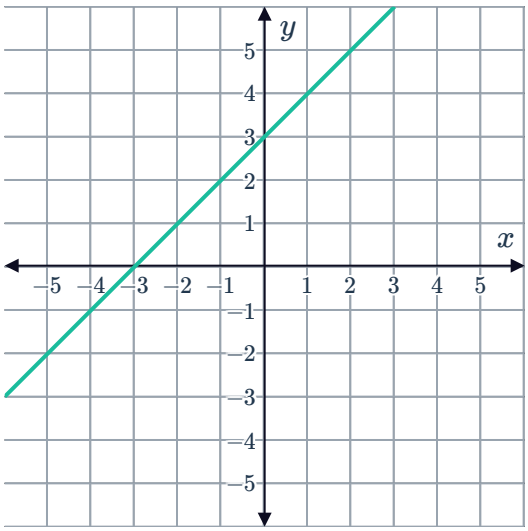
B



C

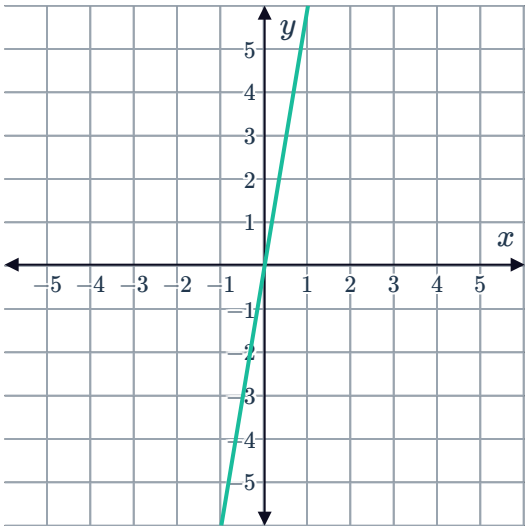


D



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E



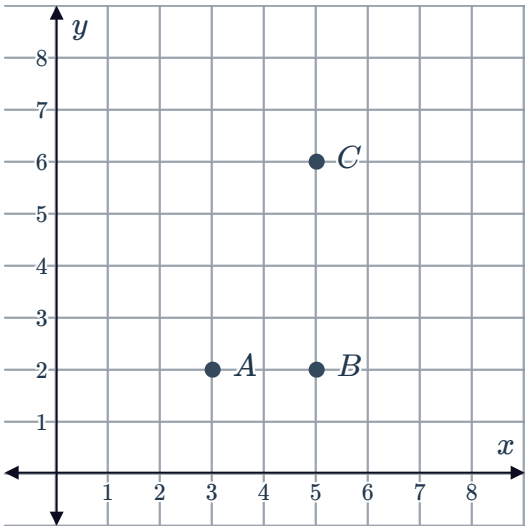
a Which line has a slope of 0?

b Which line has a slope that is undefined?

19 Consider the points on the graph.

a What is the slope of the line segment AB?

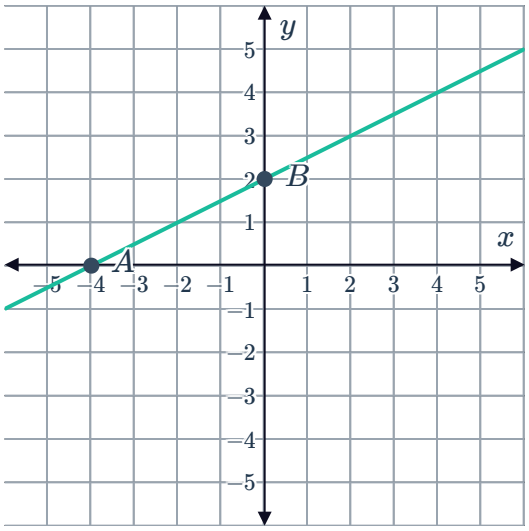
b What is the slope of the line segment BC?



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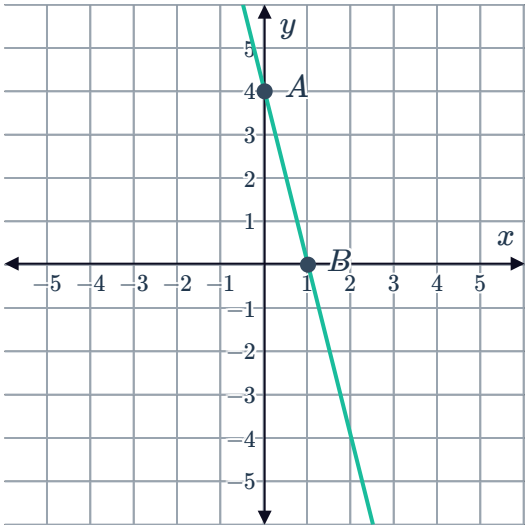
20 Consider the segment shown in the graph with Point A $(-4, 0)$ and Point B $(0, 2)$

- **a** Find the rise (change in the y value) between point A and B. Note: Ensure you have the correct sign.
- **b** Find the run (change in the x value) between point A and B. Note: Ensure you have the correct sign.
- **c** Find the slope of the segment AB.



21 Consider the segment containing Point $A\,(0, 4)$ and Point $B\,(1, 0)$

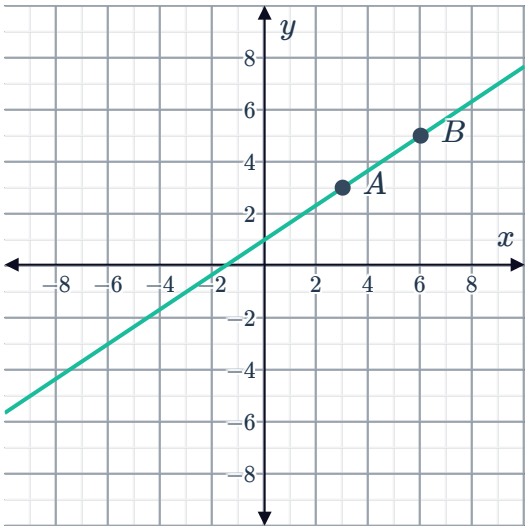
- **a** Find the rise (change in the y-value) going from point A to point B.
- **b** Find the run (change in x-value) going from point A to point B.
- **c** Find the slope of the segment AB.



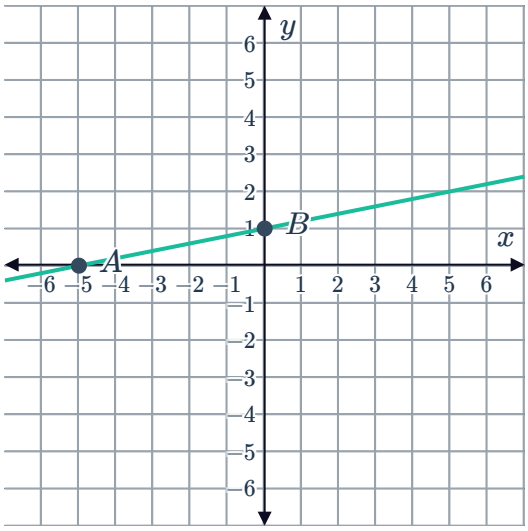
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24 What is the slope of the line shown in the graph given that:

a Point A $(3,3)$ and Point B $(6,5)$ both lie on the line.

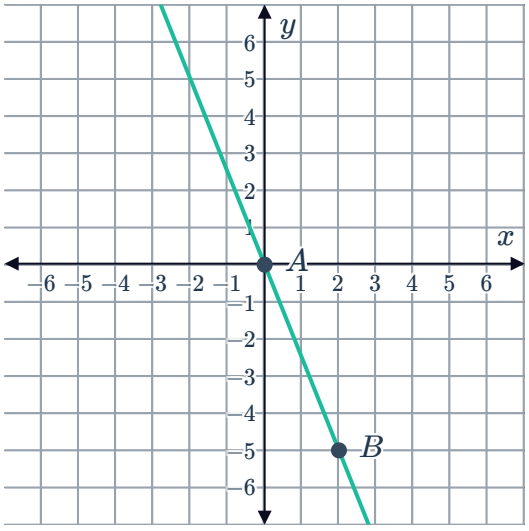


b Point A $(-5,0)$ and Point B $(0,1)$ both lie on the line.

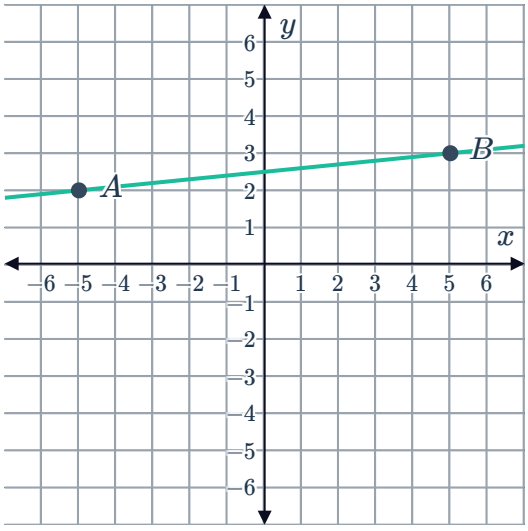


25 What is the slope of the line going through A and B?

a

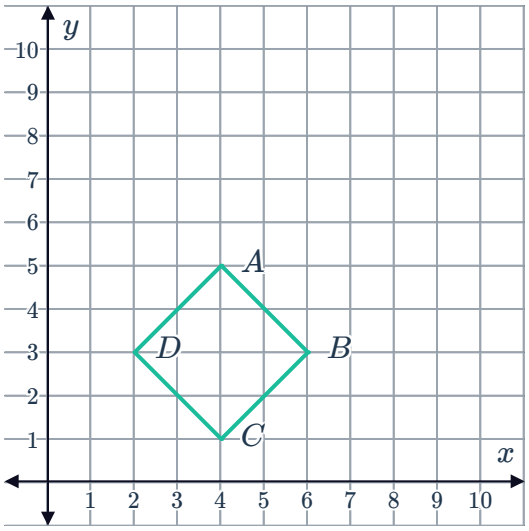


b



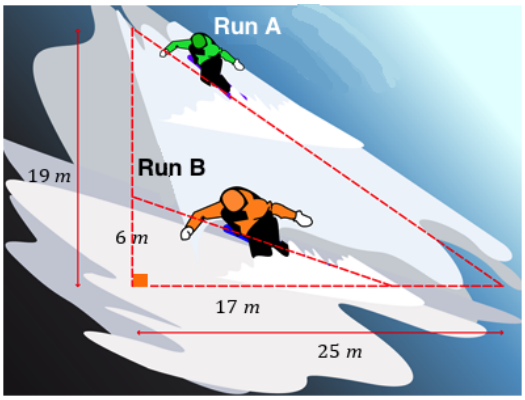
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26 The four vertices of square ABCD have been plotted on a number plane.



- **a** Find the slope of side AB.
- **b** Find the slope of side BC.
- **c** What is the product of the slope s in part (a) and (b)?

27 A certain ski resort has two ski runs.



- **a** Find the slope of ski slope A. Give your answer as a decimal to two decimal places.
- **b** Find the slope of ski run B. Give your answer as a decimal to two decimal places.
- **c** Which run is steeper, A or B?

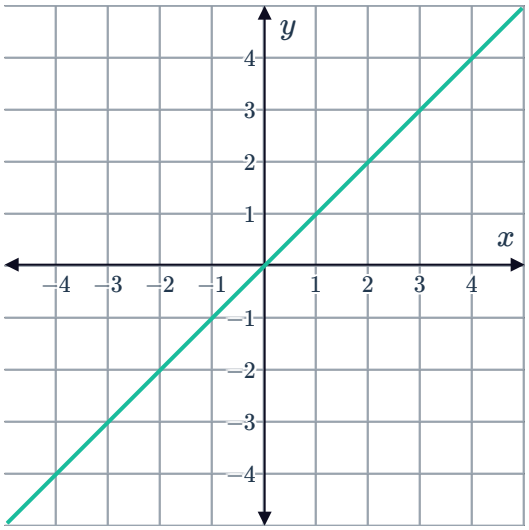
28 A staircase is to be built so that its maximum steepness is 0.6. If each step goes in 22 cm, what is the maximum height in centimeters it can rise vertically?

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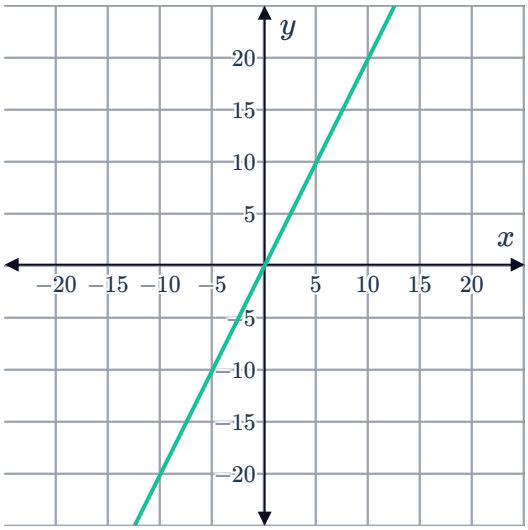
Additional questions

29 For each of the following lines, identify the slope .

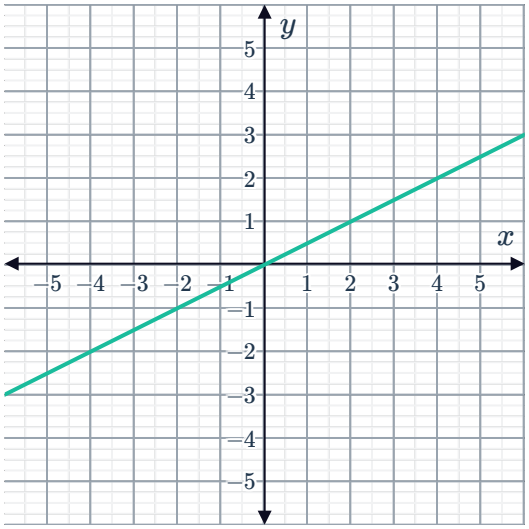
a



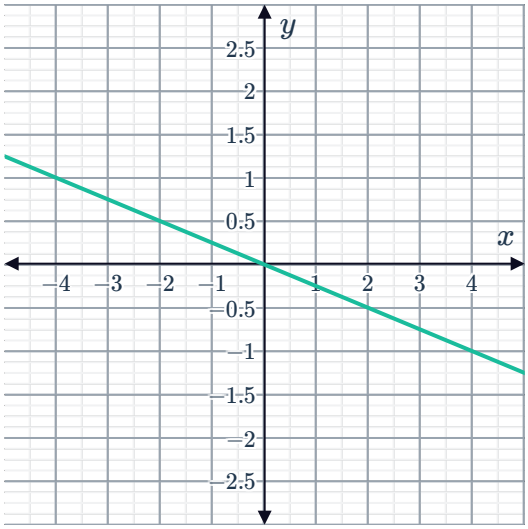
b



c

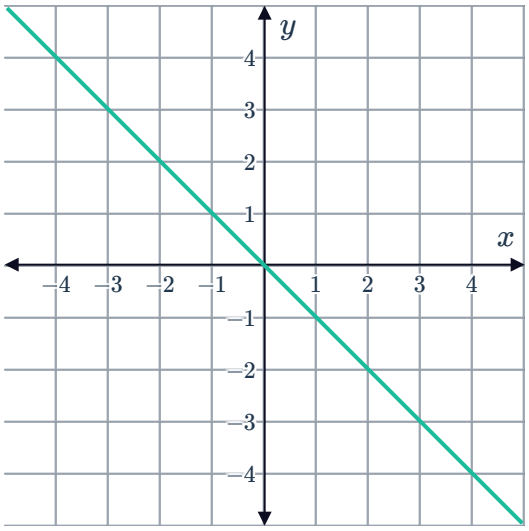


d

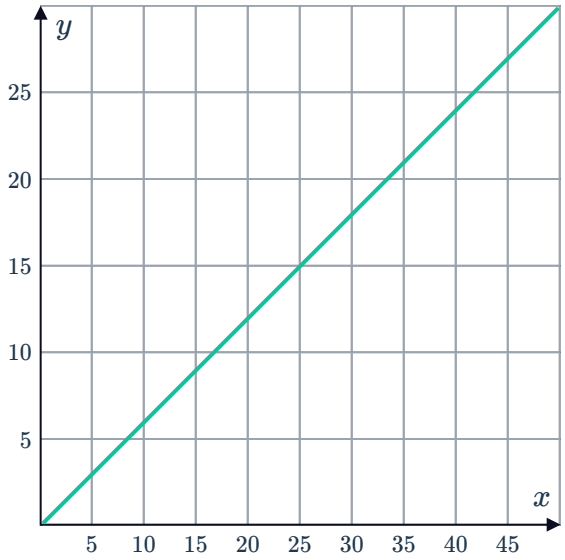


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e



f



30 For each of the following relationships, identify the slope :

a

x	y
-3	-27
-2	-18
-1	-9
0	0
1	9
2	18

b

x	y
-8	-24
-6	-18
-4	-12
-2	-6
0	0
2	6

c

x	y
0.2	1.5
0.4	3
0.6	4.5
0.8	6
1	7.5
1.2	9

d

x	y
5	-7
10	-14
15	-21
20	-28
25	-35
30	-42

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